

INFLUENCER FRAUD DETECTION USING UNSUPERVISED MACHINE LEARNING

Research Project Report

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ABSTRACT

Influencer marketing has become one of the most effective digital marketing strategies for brands and businesses. However, the increasing use of fake followers, artificial engagement, and automated bots has created significant challenges in identifying genuine influencers. Fraudulent influencer accounts can mislead businesses and result in ineffective marketing investments.

This project presents an Influencer Fraud Detection System using unsupervised machine learning techniques. A dataset consisting of 5,000 influencer profiles was created and analyzed using various engagement and growth-related metrics. Feature engineering techniques were applied to generate meaningful indicators such as Engagement Rate, Follower-Following Ratio, Comment-Like Ratio, Likes Per Post, and Comments Per Post.

To identify suspicious influencer accounts, Isolation Forest and Local Outlier Factor (LOF) algorithms were implemented. These anomaly detection techniques helped identify unusual behavioral patterns that may indicate fraudulent activities. Based on the outputs of both models, a Fraud Risk Score ranging from 0 to 100 was generated for each influencer profile.

The final results classified influencers into four categories: Safe, Review, Suspicious, and High Risk. The proposed system provides a practical solution for brands, agencies, and marketers to evaluate influencer authenticity and make informed collaboration decisions. The project demonstrates the effectiveness of machine learning in detecting influencer fraud and can be further enhanced through real-time social media data integration.

INTRODUCTION

Influencer marketing has become one of the fastest-growing forms of digital advertising. Businesses increasingly collaborate with social media influencers to promote products, services, and brands. Influencers have the ability to reach large audiences and significantly impact consumer purchasing decisions.

However, the rapid growth of influencer marketing has also led to the rise of fraudulent activities. Many influencers artificially increase their follower count, likes, comments, and engagement using bots, fake accounts, and engagement manipulation techniques. As a result, brands often struggle to identify genuine influencers and may invest marketing budgets in accounts that provide little real value.

Detecting fraudulent influencers manually is a difficult and time-consuming process due to the large volume of social media accounts and continuously changing engagement patterns. Machine learning provides an efficient solution by automatically identifying unusual behaviors and detecting suspicious profiles.

This project focuses on developing an Influencer Fraud Detection System using unsupervised machine learning techniques. The system analyzes influencer metrics such as followers, engagement rate, growth rate, likes, and comments to identify abnormal patterns. Isolation Forest and Local Outlier Factor (LOF) algorithms are applied to detect anomalies and assign a fraud risk score to each influencer profile.

The generated fraud risk score helps brands, marketing agencies, and businesses make informed decisions while selecting influencers for promotional campaigns.

OBJECTIVES

- To identify potentially fraudulent influencer profiles.
- To analyze engagement and growth-related metrics.
- To apply anomaly detection techniques using Isolation Forest and Local Outlier Factor (LOF).
- To generate a fraud risk score ranging from 0 to 100.
- To classify influencer profiles into Safe, Review, Suspicious, and High-Risk categories.
- To support businesses and marketing agencies in influencer selection and verification.

SCOPE OF THE PROJECT

This project focuses on detecting fraudulent influencer profiles using machine learning techniques. The analysis is based on influencer statistics such as followers, engagement rate, likes, comments, and growth metrics. The proposed solution can assist brands in evaluating influencer authenticity and reducing marketing risks.

DATASET DESCRIPTION

The dataset used in this project consists of 5,000 influencer profiles representing social media accounts from platforms such as Instagram and TikTok. The dataset was created to simulate real-world influencer behavior and includes both normal and potentially fraudulent profiles.

Various profile-level and engagement-related metrics were collected and engineered to support anomaly detection. The dataset contains features such as followers, following count, number of posts, average likes, average comments, growth rate, engagement rate, follower-following ratio, comment-like ratio, likes per post, and comments per post.

To improve the effectiveness of fraud detection, anomalous profiles were introduced by manipulating engagement and follower metrics. These profiles were later identified using machine learning models.

DATASET SUMMARY TABLE

Attribute	Description
Total Profiles	5000
Platform	Instagram/TikTok
Features	11+
Data Type	Numerical
Fraud Profiles	Simulated Anomalies
ML Approach	Unsupervised Learning

FEATURE ENGINEERING

Feature engineering is one of the most important steps in machine learning because it helps transform raw data into meaningful features that improve model performance. In this project, several additional features were created from the original influencer metrics to better identify suspicious and potentially fraudulent profiles.

The primary objective of feature engineering was to capture engagement behavior, audience interaction patterns, and profile credibility indicators. These engineered features provide valuable insights that may not be visible from raw follower or engagement counts alone.

The following features were created and used during the analysis:

Feature	Description
Engagement Rate	Measures audience interaction relative to followers
Follower-Following Ratio	Indicates profile credibility and influence
Comment-Like Ratio	Measures authenticity of audience engagement
Likes Per Post	Average likes received per post
Comments Per Post	Average comments received per post

ENGINEERED FEATURE EXPLANATION

Engagement Rate

Engagement Rate measures the level of interaction between an influencer and their audience. A higher engagement rate generally indicates a more active and genuine audience.

Follower-Following Ratio

This feature compares the number of followers with the number of accounts followed by an influencer. Unusual ratios may indicate suspicious account behavior.

Comment-Like Ratio

The Comment-Like Ratio helps identify whether engagement appears natural. Accounts with extremely low comments compared to likes may indicate fake engagement.

Likes Per Post and Comments Per Post

These features represent average audience interaction on content published by an influencer and help evaluate overall engagement consistency.

Feature Engineering Benefits

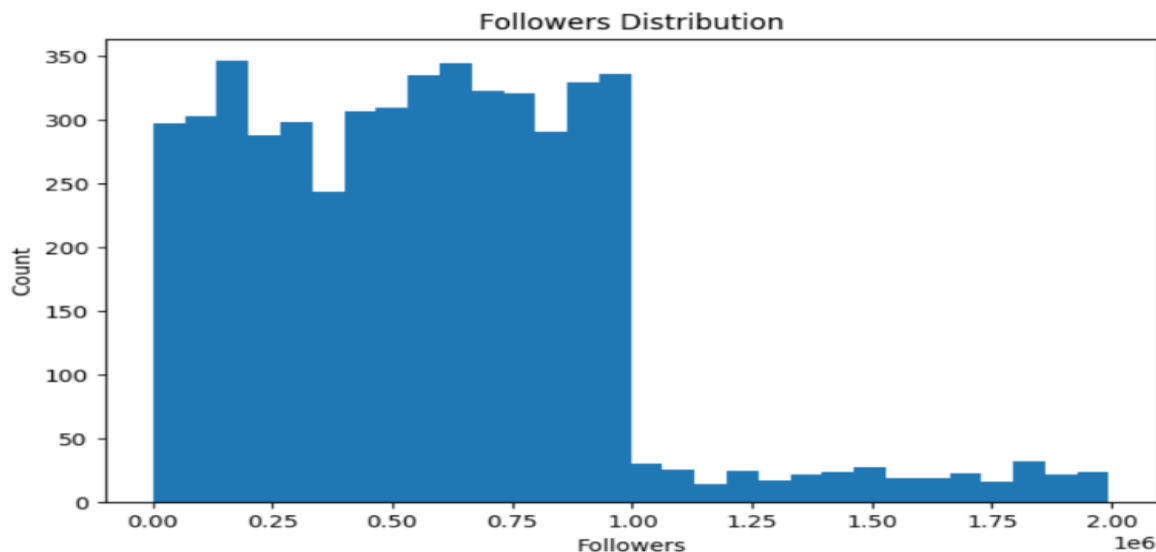
- Improved anomaly detection performance.
- Better representation of influencer behavior.
- Enhanced fraud identification capability.
- Increased effectiveness of machine learning models.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed to understand the distribution of influencer metrics and identify patterns within the dataset. EDA helps uncover trends, anomalies, and relationships between variables before applying machine learning models.

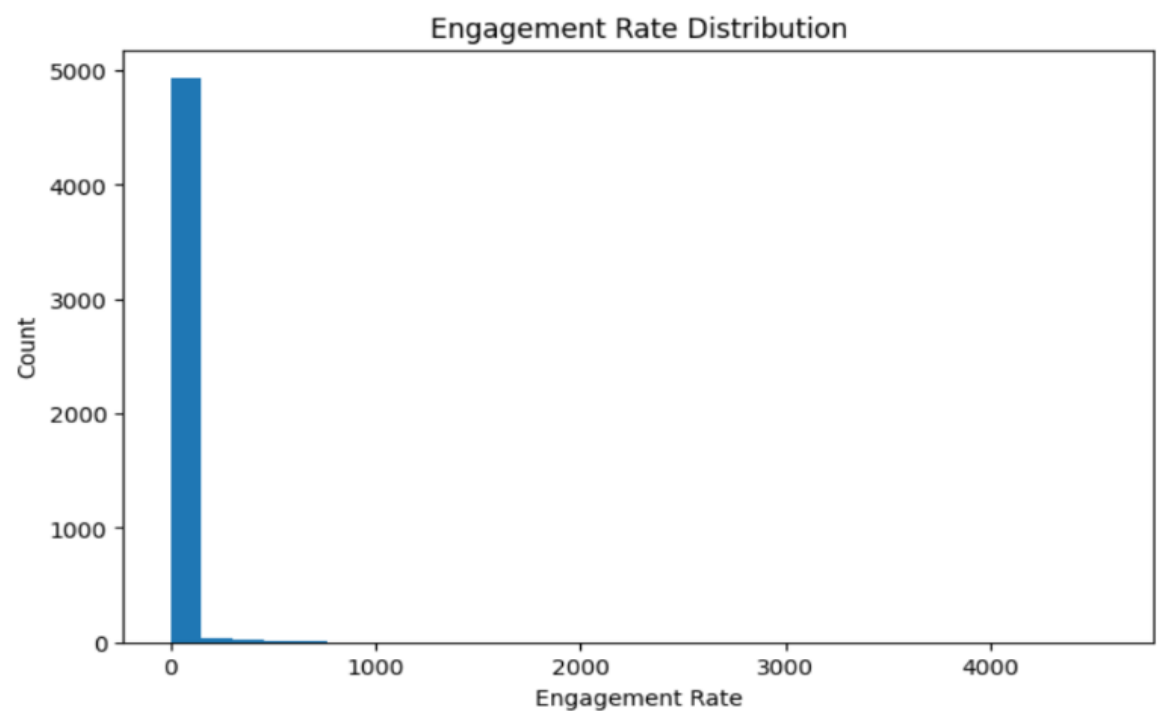
Various visualizations were generated to analyze follower distribution, engagement behavior, and growth patterns. These insights helped in understanding normal influencer behavior and identifying potentially suspicious profiles.

Figure 1: Followers Distribution of Influencer Profiles



The histogram illustrates the distribution of follower counts across influencer profiles. Most profiles fall within the low to medium follower range, while a smaller number of profiles have exceptionally high follower counts. This distribution helps identify potential outliers and understand audience reach patterns within the dataset.

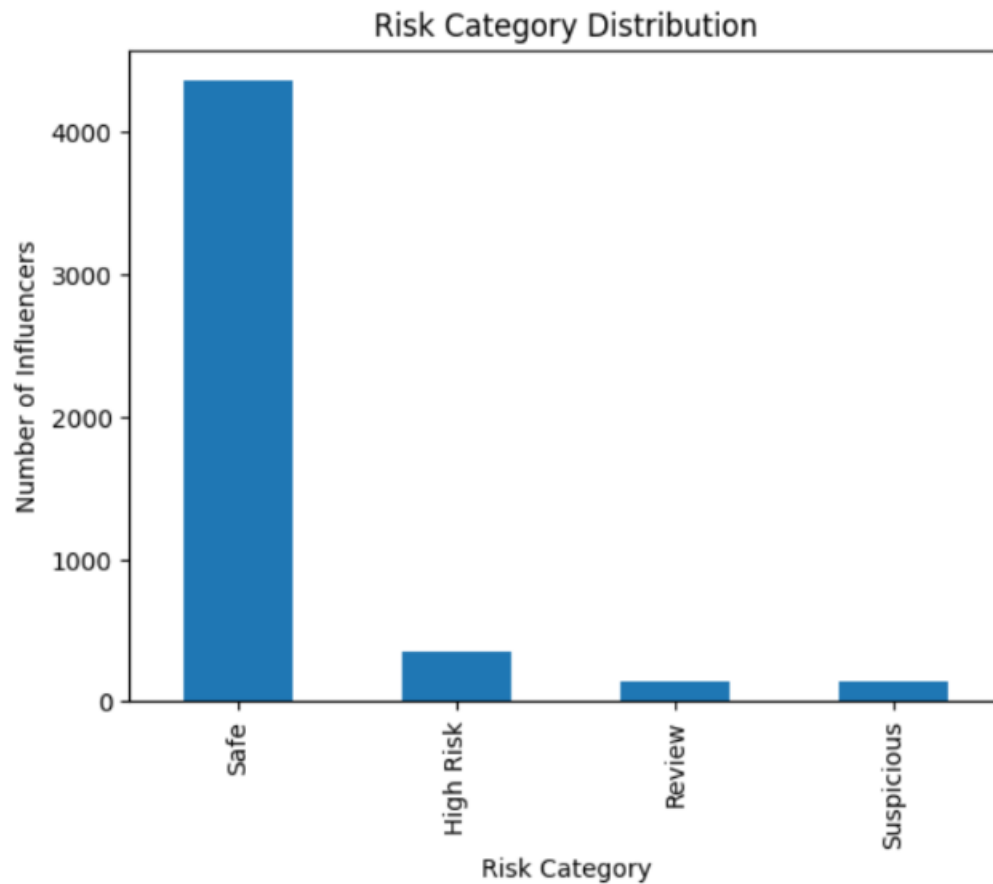
Figure 2: Engagement Rate Distribution Across Influencer Profiles



The histogram illustrates the distribution of engagement rates among influencer profiles. Most influencers exhibit engagement rates within a normal range, while a small number of profiles show unusually high engagement values. Such extreme engagement rates may indicate suspicious activity, artificial engagement, or abnormal audience behavior. This analysis helps identify potential anomalies before applying machine learning models.

RESULT AND RISK CLASSIFICATION

Figure 3: Distribution of Influencer Risk Categories



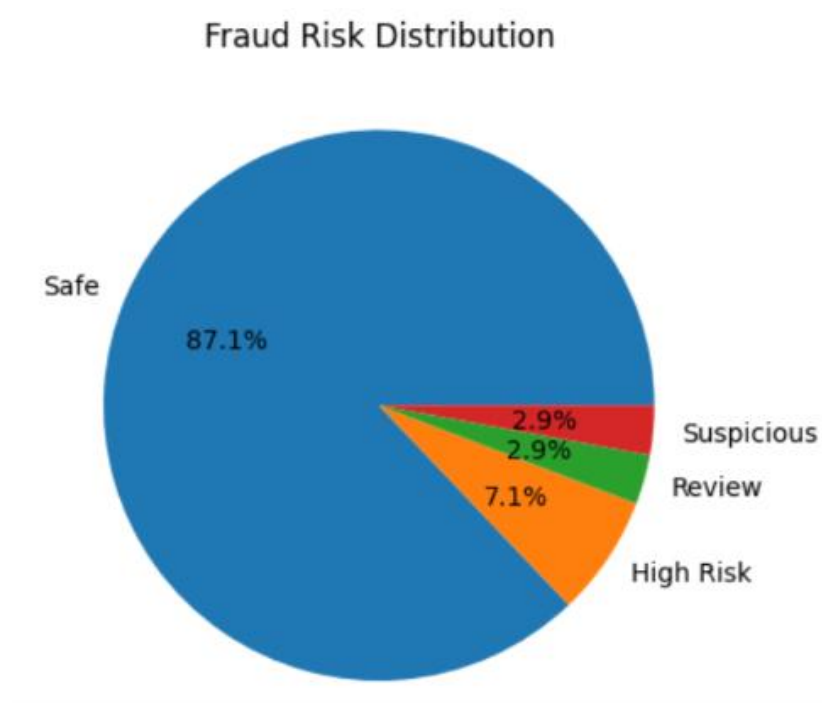
The chart presents the final distribution of influencer profiles based on the generated Fraud Risk Score. The majority of profiles were classified as Safe, indicating normal engagement behavior and authentic audience interaction. A smaller number of profiles were categorized as Review, Suspicious, and High Risk due to unusual engagement patterns and anomaly scores generated by the machine learning models. These results demonstrate the effectiveness of the proposed fraud detection framework in identifying potentially fraudulent influencer accounts.

Key Findings

- Safe Profiles: 4,355
- High-Risk Profiles: 355
- Review Profiles: 145
- Suspicious Profiles: 145

The results indicate that approximately 13% of influencer profiles required further investigation or were identified as potentially fraudulent.

Figure 4: Proportion of Influencer Risk Categories



The pie chart represents the percentage distribution of influencer profiles across different fraud risk categories. Approximately 87.1% of profiles were classified as Safe, indicating normal engagement behavior. The remaining profiles were categorized as High Risk (7.1%), Review (2.9%), and Suspicious (2.9%). These findings suggest that while most influencer accounts appear authentic, a significant minority exhibit unusual patterns that warrant further investigation.

FRAUD RISK THRESHOLDS

Risk Score Range	Category	Recommended Action
0 – 30	Safe	Approve influencer for campaigns
31 – 60	Review	Perform manual verification
61 – 80	Suspicious	Conduct detailed investigation
81 – 100	High Risk	Reject or avoid collaboration

BUSINESS DECISION FRAMEWORK

The generated fraud risk score can support brands and marketing agencies in influencer selection. Profiles categorized as Safe can be approved for campaigns with minimal review. Profiles in the Review category should undergo additional verification. Suspicious profiles require detailed investigation, while High-Risk profiles should generally be excluded from influencer marketing campaigns to reduce fraud-related risks and improve return on investment.

CONCLUSION

This project successfully developed an Influencer Fraud Detection System using unsupervised machine learning techniques. A dataset containing 5,000 influencer profiles was analyzed and enhanced through feature engineering. Isolation Forest and Local Outlier Factor (LOF) models were used to identify anomalous influencer behavior and generate fraud risk scores.

The results demonstrated that the proposed approach can effectively detect suspicious influencer profiles and assist brands in making informed influencer marketing decisions. The generated fraud risk score and risk categories provide a practical framework for influencer verification and fraud prevention.

FUTURE SCOPE

- Integration with real-time Instagram and TikTok APIs.
- Development of a live fraud monitoring dashboard.
- Use of advanced deep learning-based anomaly detection techniques.
- Automated influencer verification and recommendation systems.
- Expansion of the model using additional social media metrics.